SPRDL1-23-R-0159 (JA-00687)
Previous J&A: SPRDL1-19-R-0058 is available.

JUSTIFICATION AND APPROVAL FOR OTHER THAN FULL AND OPEN COMPETITION SPRDL1-23-R-0159

1. **Contracting Agency:** Defense Logistics Agency - Land Warren (DLA-WRN).

2. **Description of Action:** DLA-WRN proposes to award a new Firm-Fixed-Price contract to Oshkosh Defense, LLC, Commercial and Government Entity (CAGE) 75Q65, for National Stock Number (NSN) 2520-01-572-2756, Axle Assembly, Automotive, Driving, using Army Working Capital Funds.

3. **Description of Supplies/Services:** The NSN 2520-01-572-2756, part number 3824083, Axle Assembly, Automotive, Driving, is used on the Heavy Expanded Mobility Tactical Truck (HEMTT) A4. A vehicle's axle assembly connects the wheels and supports the weight of the vehicle, allowing the power generated by the engine to be transmitted to the wheels and enable vehicle movement. The deliverables for this contract will be for a base quantity of 25 each, with an option quantity of 25 each, making a total quantity of 50 each. The total maximum value of this contract is estimated at $[redacted].

Acquisition Method Code (AMC):

3 - Acquire, for the second or subsequent time, directly from the actual manufacturer.

Acquisition Method Suffix Code (AMSC):

R - The Government does not own the data or the rights to the data needed to purchase this part from additional sources. It has been determined to be uneconomical to buy the data or rights to the data. It is uneconomical to reverse engineer the part. This code is used when the Government did not initially purchase the data and/or rights. If only one source has the rights or data to manufacture this item, AMCs 3, 4, or 5 are valid. If two or more sources have the rights or data to manufacture this item, AMCs 1 or 2 are valid.

4. **Authority Cited:** The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)(1) as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. **Reason for Authority Cited:** This Automotive Axle is the 4th (rearmost) axle on the HEMTTA4 platform. Axles are a critical component of the vehicle's drive train assembly, which transfers power generated by the engine from the transmission to the wheels. The wheels are mounted to the axle assembly, providing mobility to the vehicle. The HEMTTA4 is designed and produced by Oshkosh. This axle assembly is used on many of the variants of the HEMTTA4, including the M977A4, M978A4, M985A4, M985A4GMT, M1120A4, and M1977A4, as well as some variants of the legacy HEMTTA2.

Controlled By: [redacted]
Controlled By: [redacted]
CUI Category: [redacted]
Distribution/Dissemination Control: [redacted]
POC: [redacted]

Oshkosh developed the system-level requirements for the drive train system and possesses details regarding the design intent of this component. The performance characteristics, mounting details, materials, components, specifications, and certifications of this component are unknown. Oshkosh is the only vendor who possesses the technical data and expertise to manufacture this axle.

The Government does not have rights to the Technical Data Package (TDP) or specifications to enable competitive re-procurement.

The axle is manufactured solely by Oshkosh, who owns all the data rights to the axle assembly. Only Oshkosh has the specific technical expertise and proper facilities to manufacture the axle. Additionally, Oshkosh has the experience to build the axle assemblies that allow the HEMTTA4 to perform its mission.

Oshkosh has manufactured axle assemblies that have successfully operated in the field and have met all mobility requirements outlined in the Army Technical Purchase Description. Unlike other commercially produced axles, this axle has been designed specifically for use on the HEMTTA4. Oshkosh and the Government conducted developmental and operational testing during the vehicle's design and initial production phases, which proved that Oshkosh could manufacture an axle that meets the Government's performance and reliability, availability, and maintainability (RAM) requirements. If the Government were to purchase axle assemblies that have not undergone proper verification, mission failures and potential loss of mission critical functions to the vehicle could occur. Without this proper verification, it is likely that the axle assembly would not fit or function properly, [redacted]

Due to the lack of technical data available for the axle, reverse engineering is the only recourse for the Government to increase competition through development of a competitive build-to-print TDP. The Ground Vehicle Systems Center (GVSC) Heavy Tactical Vehicle Sustainment Team has estimated the effort that would be required to reverse engineer this axle based on conversations with Subject Matter Experts in Reverse Engineering, Test & Evaluation, and Quality Engineering. The reverse engineering process consists of procuring samples of the current part, tearing down the samples to understand and characterize the assembly's components, dimensions, and materials, and designing the reverse engineered part. Several prototypes would need to be fabricated. The process to reverse engineer the part and fabricate prototypes would cost an estimated minimum of $1 million with a lead time of 12 months, of which $100,000 would be used to procure samples, $700,000 would be used to characterize the part and develop the new design, and $200,000 would be used to fabricate prototypes. In addition, the prototype would have to undergo design validation and vehicle integration testing, which would cost an additional estimated minimum of $1.5 million with an additional lead time of 24 months. The prototype testing would consist of a form, fit, and function test, a performance test, RAM testing, accelerated life testing, cold and hot environment testing, salt spray testing, and maximum load operation testing. To procure this axle competitively, a complete TDP would need to be developed, which would cost an estimated minimum of $1 million with an additional lead time of six months.

In total, reverse engineering, design validation, performance testing, and developing the TDP would require at least 42 months and an estimated minimum of $3.5 million. To neglect design validation and integration testing would be a high risk as premature failures of this item in the field could jeopardize the Army's mission in a potentially adverse operational environment. Ultimately, there is a vital need for correct and approved components to be contractually obtained under this procurement to maintain readiness and avoid vehicle dead-lining due to supply shortages.

The Government would not be able to recover, through subsequent competition, the duplication of costs involved in reverse engineering the item to create a new and competitive TDP within a reasonable amount of time. At the current average usage rate of [redacted] units each year, a current unit cost of $[redacted], and assuming 25% cost savings achieved through competition, the $[redacted] in annual savings would take over [redacted] years to offset the estimated $3.5 million cost of reverse engineering the item and developing a competitive TDP.

The stock out date for this axle was 20 June 2023. The current backorder is 26 units, and 10 vehicles are currently in Non-Mission Capable status. The 54 months required to reverse engineer this part would result in a total deficiency of 103 axles. The delay associated with reverse engineering this axle is unacceptable in fulfilling the Army's requirements and would have a significant impact on the Army's ability to sustain the HEMTTA2 fleet.

6. Efforts to Obtain Competition:

a. Effective competition: Due to the reasons cited in section 5 above, the Government is not pursuing competition for this item.

b. Subcontracting competition: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages--Subcontracting Plan";" and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

SOLE SOURCE: Notices required by FAR 5.201 will be posted.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

There are no contractors currently in the process of being qualified as an alternate source of supply for this item. Currently, there are no current or planned efforts to qualify additional sources since the Government does not possess a fully defined and validated TDP for this item.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 18 July 2023 and was open for 15 days. No responses were received.

Market research was conducted by the GVSC Heavy Tactical Vehicle Sustainment Team during August of 2023 to find suitable competition for the Oshkosh Axle Assembly. Axles are identified by characteristics such as weight rating and wheel end hardware, as well as interfaces with suspension and wheel/tire assemblies. The materials, dimensions, mounting details, and performance requirements are not available for this assembly. Due to the lack of a TDP or in-depth technical specifications of the sole-source part, the validity of researched axles is difficult to evaluate.
Other manufacturers offer similar axles, but the form, fit, and/or function of alternatives are not identical to the sole-source part. Keywords searched: axle assembly, military axle, HEMTT axle, heavy truck axle.

https://www.meritor.com/products/axles
https://www.rockwellaxles.com/rockwell_axle_specs.htm
https://spicerparts.com/parts?category=1
https://www.hendrickson-intl.com/search?ti=product&st=axle

Commercial axles offered by Meritor, Rockwell, DANA/Spicer, and Hendrickson were reviewed. There is not enough information to determine if these options will fit on the vehicle and operate in the same manner as the sole-source part. For this assembly, Oshkosh purchases a commercial axle from a supplier but modifies it for military use. The modification, which involves adding components to increase performance and properly interface with military suspensions, is proprietary to Oshkosh and is not known by the Government.

Additionally, a visual comparison of the axle from the Army technical manual to the researched alternatives was conducted, but none of the commercially available parts appear to be identical in mounting, size, or shape. Due to internal components, even a part that appeared the same externally could vary in performance, capability, or manufacturing.

When, or if, the technical information for the axle assembly is made available to the Government, additional and specific market research can be conducted, and competitive procurement can be considered.

9. Interested Sources: To date no additional sources have expressed interest in this item.

10. Other Facts: Procurement History:

Contract Number: SPRDL119D0099
Award Date: 29 May 2019
Contractor: Oshkosh Defense
Contract Value: $1,225,554.38 (Delivery orders)

Competitive Status: FAR 6.302-1 Only One Responsible Source

Contract Number: SPRDL110C0018
Award Date: 23 December 2009
Contractor: Oshkosh Corp
Contract Value: $1,322,250
Competitive Status: Unknown; contract file non-existent

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [redacted]
Title: Division Chief – Sustainment Engineering
Email: [redacted]
Signature:

10/19/2023

X [redacted]

Signed by: USA

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [redacted]
Title: Division Chief
Email: [redacted]
Signature:

10/24/2023

X [redacted]

Signed by: [redacted]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and/ or price analysis and a comparison to historical purchases. The contractor will not be required to submit certified cost and pricing data as this has been determined to be a commercial product.

Name: Reinbolt, Adam R.
Title: Contracting Officer
Email: adam.reinbolt@dla.mil
Signature:

10/20/2023

X Adam Reinbolt

Signed by: REINBOLT.ADAM.ROSS.1294337225

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Reinbolt, Adam R.
Title: Contracting Officer
Email: adam.reinbolt@dla.mil
Signature:

10/20/2023

X Adam Reinbolt

Signed by: REINBOLT.ADAM.ROSS.1294337225

Approval Authority for Acquisitions Greater Than $750,000 but Less Than $15M

Based on the foregoing justification, I hereby approve the procurement of the following:

a. Noun: Axle Assembly, Automotive

b. NSN: 2520-01-572-2756

c. Quantity: 25 each

d. Option Quantity: 25 each

e. Estimated Total Value: $[redacted]

This action will be awarded as other than full and open competition pursuant to the authority of 10 United States Code (U.S.C.) 3204(a)(1) and subject to the availability of funds. This approval is subject to the provision that the property herein described has otherwise been authorized for acquisition.

Signature:

X DAPO.DOUGLAS.A.1230220014 Digitally signed by DAPO.DOUGLAS.A.1230220014 Date: 2023.10.26 16:37:41 -04'00'